Willian David Correa Valencia

Systems Engineer | Backend, IoT and Cloud

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Summary

Systems Engineer (Universidad del Valle, 2025), registered with COPNIA, the Colombian engineering council. I work where software meets physical infrastructure. Today I audit public-lighting networks in the field, combining data analysis, GIS tools and technical reporting under regulation. My technical focus is backend development in Python, SQL and REST APIs, backed by training in machine learning, generative AI (RAG and vector databases), IoT and LoRaWAN, and AWS. I am building toward a cloud engineering role and am open to remote work with international teams and to relocation.

Professional Experience

Systems Engineer, Technical Oversight *2025 – Present*
INTECOAL S.A.S., public-lighting oversight and consulting

- Audit public-lighting infrastructure in several municipalities for compliance with RETILAP, the Colombian lighting regulation, including the Tuluá lighting contract: installation, maintenance and luminaire condition.
- Survey and geo-reference electrical assets with GIS tools, building a consolidated asset inventory for follow-up and audits.
- Write technical oversight reports and verification records for government entities.
- Analyze consumption and network-status data to detect billing and maintenance inconsistencies.
- Provide IT support and systems maintenance for the company, including Microsoft 365 administration.

Technical Assistant and Software Support *2024*
Globalatam

- Installed, configured and repaired software on automotive diagnostic machinery (wheel aligners, sensors).
- Gathered technical requirements and wrote documentation following ISO/IEC/IEEE 29148:2018 and ISO 24641:2023.
- Supported clients directly and built corporate websites, working through remote-access tools.

Technical Skills

- **Languages:** Python, SQL, JavaScript, TypeScript.
- **Backend and APIs:** FastAPI, Django REST, NestJS, Next.js, Spring Boot, REST design, JWT authentication.
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, SQLite.
- **Generative AI:** vector databases (pgvector, ChromaDB, MongoDB, Cassandra), embeddings, similarity search, Retrieval-Augmented Generation (RAG), LangChain.
- **Machine Learning:** TensorFlow, scikit-learn, Keras; evaluation with F1, precision and recall.
- **GIS and IoT:** ArcGIS, asset geo-referencing, LoRaWAN, IoT (HCIA-IoT), AutoCAD 2D/3D.
- **IT and systems:** Microsoft 365 administration, systems maintenance, technical support.
- **Cloud and DevOps:** AWS (S3, Lambda, Amplify), Docker, Git, GitHub Actions, Nginx, Railway, Vercel.

Projects

TallerTech: auto repair shop management system *2024*

- Full-stack app with a FastAPI backend, React frontend and PostgreSQL database. JWT authentication, CRUD operations, live charts, deployed on Vercel and Railway.
- taller-tech.vercel.app | GitHub

Andean instrument identification with AI (undergraduate thesis, published) *2024*

- Classified instrument audio with convolutional neural networks, MFCC features and Mel spectrograms. Python audio pipeline integrated with Django and React.

Rackituq: a programming language built from scratch *2025*

- Designed and implemented the grammar, interpreter and evaluator in Racket, applying language theory, parsing and interpreter design.
- github.com/FilosofoCaucano/rackituq

Camiones del Sur: layered-architecture simulator *2026*

- Road-transport simulator in Python (ursina). The domain logic is pure and testable without the graphical interface.
- github.com/FilosofoCaucano/camiones-del-sur

Also: Cenit, a desktop and Android app that finds when the sun crosses the zenith over any Colombian municipality (GitHub); a reinforcement-learning agent for the board game Linja (2023); Colombia se puebla, a visual timeline of the settlement history of Colombia (2026).

Education

Systems Engineering *2020 – 2025*
Universidad del Valle, Tuluá campus — Graduated with Academic Excellence

- Research area: Machine Learning and Intelligent Systems. Thesis advisor: Prof. Héctor Fabio Ocampo.

Certifications and courses

- Vector Database Fundamentals (4-course specialization), IBM / Coursera *(2026)*
- AWS Cloud Technical Essentials, Coursera *(2026)*
- Meta Backend Developer, Coursera *(2026)*
- Introduction to LoRaWAN, Semtech *(2026)*
- HCIA-IoT, Huawei *(2026)*
- AutoCAD 2D and 3D, SENA *(2026, 80 hours)*
- Relational Databases, Coursera *(2024, grade 86/100)*
- CONTE electrical competency certification *(in progress, expected 2027)*

Languages and Publication

- **Languages:** Spanish (native), English (upper-intermediate, B2+), Portuguese (basic).
- **Publication:** Correa Valencia, W. D. *Identification of Andean instruments using Machine Learning techniques* (in Spanish). Universidad del Valle, 2024.